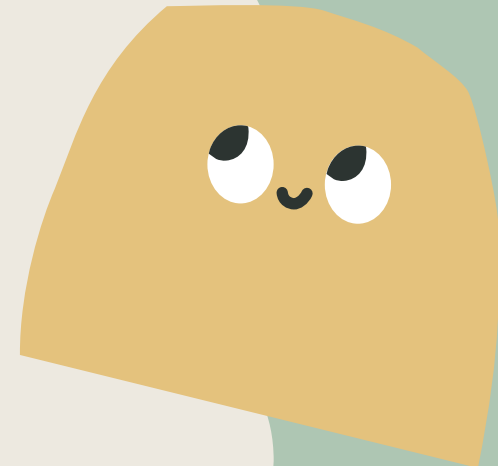
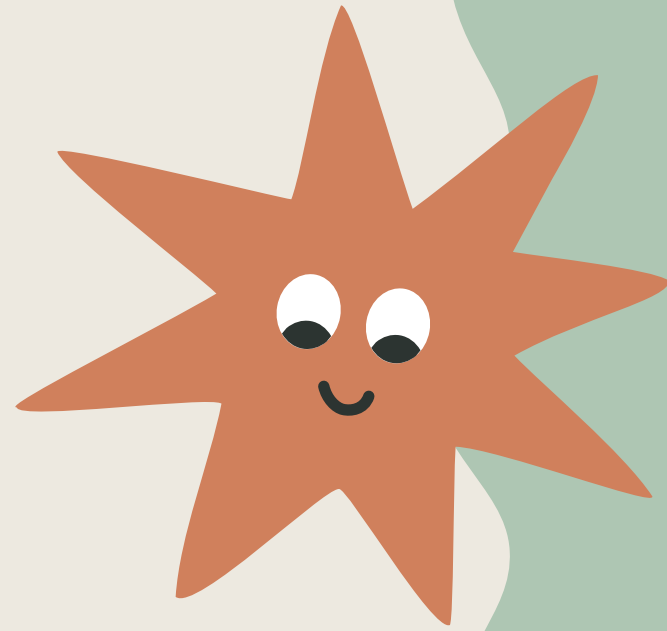


Doubly Linked List

Lecture 12



Beginner Level Questions

- Q. What is an array? Write two advantages and two limitations of using arrays.
- Q. What is traversal in an array? Write an algorithm to traverse an array.
- Q. What is the difference between insertion and deletion in an array?
- Q. What is a singly linked list? Explain the structure of a node with a diagram.
- Q. What is the purpose of the next pointer in a singly linked-list node?
- Q. What is the difference between an array and a singly linked list?
- Q. What is a circular linked list? How is it different from a singly linked list?

Beginner Level Questions

Q. Compare singly, circular, and doubly linked lists with respect to their node structure.

Q. What is the time complexity of the following operations?

- Array traversal
- Array search
- Linked-list traversal
- Linked-list search
- Insertion at the beginning of a linked list
- Deletion from the beginning of a linked list

Beginner Level Questions

Q. Write a C++ program to traverse an integer array and display all its elements.

Q. Write a C++ function to traverse a circular linked list.

Q. Write a C++ function to traverse a doubly linked list in the forward direction.

Q. Write a C++ program to delete an element from a particular index of an array.

Q. Write a C++ program to create a singly linked list containing five nodes and display all nodes.

Intermediate Level Questions

Q. Explain the three cases of insertion in an array:

- Insertion at beginning
- Insertion at end
- Insertion at a particular index

➤ Write the algorithm for each case.

Q. Explain the three cases of deletion in an array:

- Deletion from beginning
- Deletion from end
- Deletion from a particular index

➤ Write the algorithm for each case.

Intermediate Level Questions

Q. Why does insertion at the beginning of an array generally require more operations than insertion at the end?

Q. Write an algorithm to insert a new node at the end of a singly linked list

Q. Explain forward and backward traversal in a doubly linked list

Q. Why can a doubly linked list be traversed in both directions while a singly linked list cannot?

Intermediate Level Questions

- Q. Write a C++ function to insert an element at a particular index in an array
- Q. Write a C++ function to find the largest and smallest elements of an array
- Q. Write a C++ program to count the number of occurrences of a given value in an array
- Q. Write a C++ program to count the number of nodes in a singly linked list
- Q. Write a C++ function to delete the last node of a circular linked list.
- Q. Write a C++ function to traverse a doubly linked list in the backward direction **without using a tail pointer**.

Advanced Level Questions

Q. Consider: $10 \rightarrow 20 \rightarrow 30 \rightarrow 40 \rightarrow \text{NULL}$

Write the exact pointer changes required to delete node 30

Q. A student implements a singly linked list and claims that insertion at the end is always $O(1)$. Is the claim correct if the list contains only a head pointer? Justify your answer

Q. Consider the following doubly linked list:

$\text{NULL} \leftarrow 10 \rightleftarrows 20 \rightleftarrows 30 \rightleftarrows 40 \rightarrow \text{NULL}$

Write all pointer changes required to insert 25 between 20 and 30

Advanced Level Questions

- Q.** A programmer uses a doubly linked list but forgets to update the prev pointer during insertion. What problems can occur during backward traversal?
- Q.** Write a C++ function that reverses a singly linked list
- Q.** Write a C++ function to determine whether a singly linked list contains duplicate values
- Q.** Write a C++ function to find the middle node of a singly linked list
- Q.** Write a C++ function to merge two sorted singly linked lists into one sorted linked list

Advanced Level Questions

Q. Write a C++ code to remove duplicate elements from a singly linked list

Q. Write a C++ program that converts an array into a singly linked list

Q. Write a C++ program that copies all elements of a singly linked list into an array

Q. A university wants to maintain a student record system where records are frequently inserted and deleted. Explain whether you would choose an array, singly linked list, circular linked list, or doubly linked list. Justify your choice based on the required operations